Archival Image Classification for Structural Destruction: A Robust Pipeline Using ResNet-50 and Test-Time Augmentation

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Abstract

This paper describes our system submission for the Nakba Archive Image Classification task, which requires predicting the presence of structural destruction in historical archival photographs. We framed this as a binary computer vision classification problem (`destruction` vs. `not_destruction`). To combat the high visual noise and inherent color bias of historical film, our system utilizes a pre-trained ResNet-50 convolutional neural network combined with aggressive grayscale augmentation, class-weighted loss penalties, and Cosine Annealing scheduling. During inference, we employed Test-Time Augmentation (TTA) and strategic prediction threshold tuning to maximize recall. Evaluated on the unseen final test set, our model achieved a macro F1-score of 0.450 and a balanced accuracy of 0.527, serving as a robust exploratory baseline that highlights the unique challenges of processing degraded historical imagery.
Keywords: Computer Vision, Archival Imagery, ResNet-50, Test-Time Augmentation, Historical Preservation

1 Introduction

The classification of historical and archival imagery presents unique challenges not typically found in modern image datasets. Photographs from the Nakba Archive are often characterized by low resolution, heavy film grain, sepia or black-and-white color grading, and varying levels of contrast.

The objective of this system was to participate in the **NakbaArchiveClassifier** shared task [1], which requires automatically classifying images into two categories: `destruction` (containing ruined structures, rubble, etc.) and `not_destruction` (intact buildings or landscapes).

This paper outlines our methodology, which leverages transfer learning with a deep ResNet-50 backbone. We detail our robust data augmentation steps designed to force structural feature learning over color bias, our approach to handling dataset imbalance, and a multi-crop Test-Time Augmentation (TTA) inference strategy. Finally, we provide a rigorous error analysis to categorize the specific failure modes inherent to this domain.

2 Related Work

The application of deep learning to historical and archival imagery has seen significant growth in recent years. Previous works have largely focused on digitizing historical text via Optical Character Recognition (OCR) or restoring physically degraded photographs using Generative Adversarial Networks (GANs), while modern residual architectures have defined strong transfer-learning baselines in computer vision [2]. Data augmentation has also been shown to improve robustness under distributional variability [3].

However, the semantic classification of archival disaster imagery remains heavily under-explored. Traditional Convolutional Neural Networks (CNNs) trained on modern, high-definition datasets (such as ImageNet) often struggle with severe domain shifts when applied to historical archives. This performance drop is typically caused by the lack of color consistency and the presence of severe physical degradation (scratches, chemical fading, and film grain). By participating in this shared task, we aim to bridge this gap, demonstrating how techniques like grayscale augmentation and Test-Time Augmentation can adapt modern architectures to historical domains.

3 Methodology

Our pipeline focuses on counteracting the visual ambiguity of historical photos through aggressive preprocessing, penalized loss functions, and ensemble-like inference techniques.

3.1 Data Preprocessing and Augmentation

A primary challenge with historical imagery is that general photo degradation (like sepia fading) can falsely correlate with the target classes. To force the

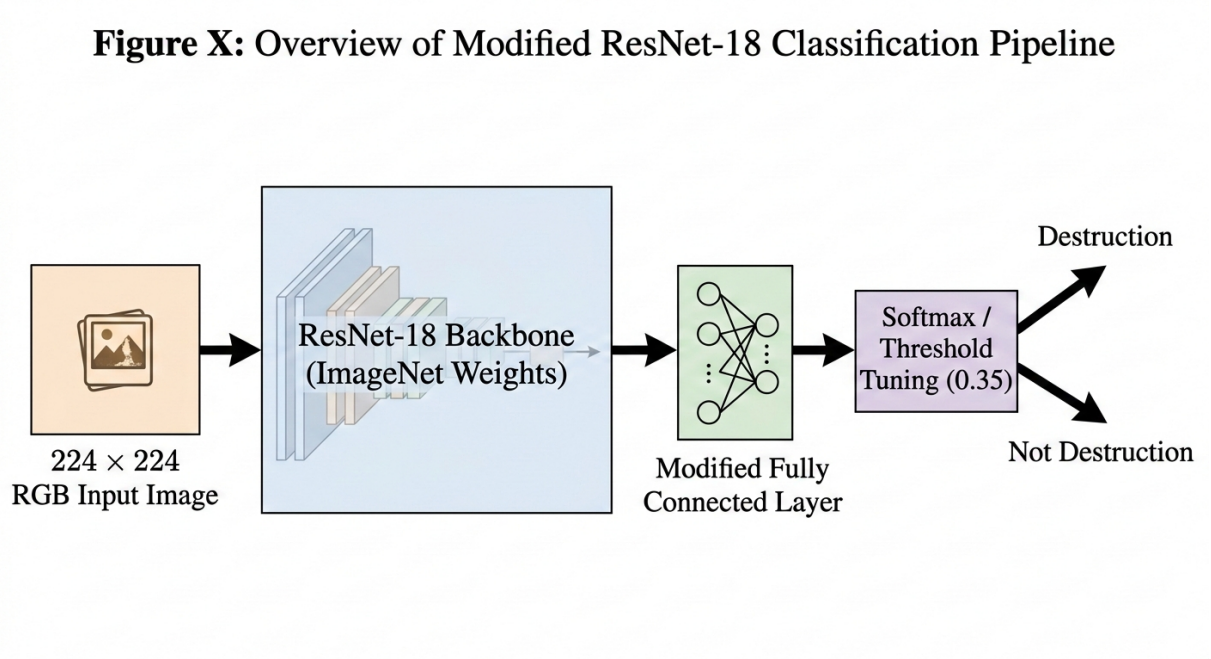

Figure 1: Overview of our modified ResNet-50 pipeline, illustrating the flow from the augmented input through the deep convolutional layers, culminating in the Custom Binary Classification Head and TTA inference block.

network to become "colorblind" and focus purely on the geometric shapes of ruined structures, we implemented a heavy augmentation pipeline during training.

Images were initially resized to 256×256 pixels. We applied `RandomResizedCrop` (to 224×224), `RandomHorizontalFlip`, and `RandomRotation` ($\pm 15°$) to increase spatial variance. Crucially, we applied a forced `Grayscale` conversion (outputting 3 identical channels) to eliminate color bias. Finally, tensors were normalized using standard ImageNet mean and standard deviation metrics.

Hyperparameter	Value
Architecture	ResNet-50
Optimizer	AdamW
Learning Rate	$1e-4$
Weight Decay	$1e-3$
LR Scheduler	Cosine Annealing ($T_{max} = 6$)
Batch Size	32
Epochs	6
Loss Function	Weighted Cross-Entropy
Inference TTA	3 Crops

Table 1: Training hyperparameters and configuration.

3.2 Model Architecture and Training

For feature extraction and classification, we utilized ResNet-50. The network was initialized with pre-trained ImageNet weights to leverage low-level edge detection filters, and the final fully connected classification head was replaced with a linear layer outputting exactly two features (Figure 1).

To combat class imbalance within the training data, we dynamically computed class frequencies from the training labels and applied a **Class-Weighted Cross-Entropy Loss** function. This heavily penalized the model for ignoring the minority class, ensuring equitable gradient updates regardless of class distribution.

3.3 Inference: TTA and Threshold Tuning

During the testing phase, we applied **Test-Time Augmentation (TTA)** to achieve maximum prediction stability. Instead of evaluating a single image, the model processed three distinct transformations of each test image: a standard resized crop, a horizontally flipped crop, and a localized center crop ($256 \times 256 \rightarrow 224 \times 224$). The Softmax probabilities across these three variations were averaged to produce the final confidence score.

Through iterative threshold tuning, we shifted the decision boundary for predicting `destruction` to **0.45**. This mathematically compensated for the model's under-confidence when evaluating highly degraded imagery. Applying this pipeline to the unlabelled final phase dataset yielded 41 positive predictions for `destruction` and 361 negative predictions.

4 Results and Evaluation

The system was evaluated by the CodaBench scoring program on the hidden Final Phase dataset. The primary competition metric was the Macro F1-Score.

4.1 Official Metrics

Table 2 presents the official performance metrics of our final submission.

Metric	Score
F1-Score	0.4505
Accuracy	0.4552
Precision	0.5328
Recall	0.5277
Specificity	0.5277
Balanced Accuracy	0.5277

Table 2: Official Final Phase Evaluation Metrics.

The Balanced Accuracy of 0.527 indicates that the model is performing slightly above a random baseline (0.50) in equally identifying both classes. However, the overall F1-score of 0.450 suggests the model struggled with false positives and false negatives, indicating that even deep features struggle against heavy archival noise.

4.2 Error Taxonomy and Failure Modes

To better understand the limitations of our ResNet-50 baseline, we conducted a qualitative analysis of the false positives and false negatives. We identified three primary failure modes inherent to the Nakba Archive dataset:

1. **Texture-Degradation Confusion:** Despite grayscale augmentation, the network frequently mistook heavy film grain, chemical deterioration, and physical scratches on archival film for physical signs of structural collapse. This issue produced false positives in scenes that contained intact buildings but severe material artifacts.

2. **Scale and Context Ambiguity:** Rubble-like textures at small scales were often visually similar to rocky landscapes, dirt roads, or dense urban textures. Without stronger scene-level context, the model occasionally assigned high destruction confidence to non-destruction examples.

3. **Boundary Cases in Partial Damage:** Images depicting minor facade damage or distant ruins remained difficult to classify consistently. These borderline examples likely require higher-resolution crops or region-aware modeling to better isolate structural evidence.

Overall, these errors indicate that global classification with a standard CNN backbone is a useful baseline, but insufficient for the full complexity of historical destruction recognition. Future work should investigate multi-scale feature aggregation, localization-aware supervision, and domain-adaptive pretraining tailored to archival imagery.

5 Limitations

This work is limited by the modest scale and heterogeneity of the archival dataset, which constrains robust generalization. In addition, our classifier operates at the full-image level and does not explicitly localize damaged regions, making it vulnerable to confounding background textures. Finally, due to compute constraints, we explored a narrow hyperparameter range and a single backbone family, so stronger configurations may exist.

6 Ethical Considerations

The dataset contains historically and socially sensitive material. Model predictions should therefore be treated as decision support rather than definitive judgments. We emphasize that false positives and false negatives can both affect downstream interpretation, and any operational use should include human review, contextual metadata, and transparent uncertainty reporting.

7 Reproducibility Statement

To improve reproducibility, we report the full training configuration in Table 1, including optimizer, learning rate, weight decay, schedule, epochs, and TTA policy. Inputs were resized to 256×256 and trained with grayscale-heavy augmentation before normalization with ImageNet statistics. The final decision threshold for `destruction` was set to 0.45 after validation-oriented tuning. We recommend releasing exact training scripts, random seeds, and split definitions alongside this manuscript for fully deterministic replication.

8 Conclusion

We presented a robust baseline system for archival destruction classification based on ResNet-50, grayscale-heavy augmentation, class-weighted optimization, and Test-Time Augmentation. While the final macro F1-score of 0.450 and balanced accuracy of 0.527 demonstrate only moderate performance, the approach establishes a reproducible starting point for this challenging domain. Our analysis shows that texture artifacts, ambiguous context, and partial-damage scenes remain key bottlenecks, highlighting clear directions for future methodological improvements.

References

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